

SYED MUHAMMAD HANZALA BIN AHSAN

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EDUCATION

BS Computer Science | FAST - National University of Computer and Emerging Sciences

2023 - Expected 2027

Lahore, Pakistan

CGPA: 3.93/4.00 | Six semesters completed | Dean's List (Semesters 1-5) | Gold Medalist and 1st Position (Fall 2025)

EXPERIENCE

AI/ML Team Lead | Özyeğin University

Jun 2026 - Aug 2026

Remote - Istanbul, Türkiye

- Led the annual-filings RAG workstream, owned the shared Git repository, coordinated implementation across the team, and served as AWS architecture lead for a production-oriented financial intelligence chatbot.
- Designed an end-to-end pipeline for 158,570 SEC 10-K chunks: BGE embeddings, PostgreSQL/pgvector cosine HNSW retrieval, multi-query decomposition, cross-encoder reranking, requirement-aware evidence selection, and grounded LLM generation with citations.
- Architected a cost-conscious AWS deployment using Amazon Bedrock Nova models, a GPU inference service on EC2 g4dn.xlarge, cross-region VPC peering to the database host, private routing, and automated instance shutdown controls.

Lead Software Developer | BookJane

May 2026 - Aug 2026

Remote - Canada

- Led development and deployment of an internal HRIS using Next.js 16, TypeScript, Tailwind CSS, PostgreSQL (Neon), Prisma 7, Auth.js v5, Google OAuth 2.0, and Vercel CI/CD.
- Implemented server-enforced RBAC, manager/owner authorization and cycle prevention, database-backed sessions with immediate offboarding revocation, just-in-time user provisioning, and immutable audit logs for sensitive HR actions.
- Built leave workflows with accrued/used/pending balance states, approval-aware accounting, employee invitation and notification emails through Nodemailer, and type-safe API routes and server components.

SELECTED PROJECTS

Retail Intelligence Annual Filings RAG | Python, PyTorch, BGE, pgvector, AWS Bedrock/EC2

Summer 2026

- Engineered a citation-grounded RAG pipeline over 158,570 SEC 10-K chunks using BGE embeddings, pgvector HNSW retrieval, query decomposition, cross-encoder reranking, requirement-aware evidence allocation, and Amazon Nova generation.
- Evaluated the frozen pipeline on unseen questions: 88/100 completed successfully; targeted retrieval optimization raised seven-miss audit Hit@5 to 1.00 and Recall@5 to 0.958 without rebuilding embeddings.

YouTube Intelligent Video Analysis | PyTorch, RoBERTa, ResNet-50, LightGBM, FastAPI

Spring 2026

- Developed a five-module pipeline for violence detection, age classification, seven-class harm analysis, 16-class category prediction, and an explainable 0-100 impact score from YouTube metadata, transcripts, and sampled frames.
- Applied hierarchical RoBERTa with sliding-window attention to full transcripts and ResNet-50 transfer learning to ~3,000 violence clips; achieved 0.958 ROC-AUC for violence detection and 0.683 macro-F1 for age classification.

Echo - Encrypted LAN Chat | Java, Spring Boot, React, WebSockets, FastAPI, Docker

Fall 2025

- Built an end-to-end encrypted real-time LAN messenger for text, audio, and images using Spring Boot WebSockets, React, PostgreSQL, and Flyway migrations.
- Designed five transformer-based Python/FastAPI microservices for translation, summarization, toxicity detection, smart replies, and deadline extraction; containerized the frontend, backend, and AI services with Docker Compose.

TECHNICAL SKILLS

AI & Data: Python, PyTorch, scikit-learn, Hugging Face Transformers, NLP, RAG, embeddings, vector search, cross-encoders, computer vision, XGBoost

Cloud & Backend: AWS (Bedrock, EC2, VPC), FastAPI, Next.js API routes, Node.js, Spring Boot, Docker, Vercel, REST APIs, CI/CD

Data & Web: PostgreSQL, pgvector, Prisma ORM, SQL, React, Next.js, TypeScript, Tailwind CSS, Git/GitHub, Java, C/C++